Performance Measures of Agility for Mobile Robots

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Categories and Subject Descriptors

C.4 [**PERFORMANCE OF SYSTEMS**]: Modeling techniques

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1. INTRODUCTION

Agility refers to a system's ability to change some aspect of its current situation. For example, consider *agile manufacturing* which concerns the ability to *change a particular manufacturing setup* from one product to another without much downtime [3, 8, 25]. In aeronautics, several scalar metrics have been proposed for *aircraft agility* which together measure an aircraft's ability to *change its speed and direction* [21, 1, 2].

In robotics, it is difficult to find a definition of agility or any scalar metrics for measuring it, although several works refer to it. This is true for all robots and is especially true for legged robots. The proposed definitions are often more implicit than explicit, but all discuss the robot's ability to change some aspect of its situation or state. In examining the field of legged robots, it is claimed that the robot's agility stems from its mechanical design, control, or both, as summarized in the following two paragraphs.

Design: Several works claim agility by design, including [17] because the multi-legged, segmented robot can change its overall body configuration allowing it to follow the terrain's contours. An innovative leg design is discussed in [19] which is attributed to the hexapod robot's agility discussed in terms of its speed and its ability to climb obstacles. Similar work discusses the effect of actuation efficiency on agility in terms a multi-legged robot's speed and ability to climb over obstacles [18]. In [16] an actuation scheme that allows a biped to perform acrobatic movements, exemplifying agility, is explored. In [10] an analysis of the robots ability to change the configuration of its legs was considered an indicator of agility. A quadruped climbing robot in [24] is claimed to be agile because of its ability to climb and change direction. A turning criteria, or direction change, at a fairly high speed is used in [15] for examining the agility of a wall climbing robot. The design of a climbing robot in [22] also mentions agility where the design is evaluated mainly in terms of its speed. Other design studies discuss several issues, speed being a central one, but it is unclear how agility was quantified [12, 11]. The work in [14] analyzes acceleration capability as a measure of agility, similar to what is used herein, except that a polytope characterization is used.

Control: In [7] agility in a hexapod micro-robot is obtained through fine tuning the control to increase its speed and ability to change direction. A review of monopod robots [23] mentions agility only in the abstract, but it has been claimed that monopods are agile because their hopping motion allows them to change directions at each ground contact and to perform acrobatic movements [20]. A controller for executing a self-righting motion, which could be considered acrobatic or agile, was developed in [9]. In [13] a genetic algorithm was used to evolve a controller to execute *dynamic maneuvers* consisting of high-speed turns and a running jump.

Nearly all of the works discussed in the "Design" and "Control" sections above include the word "agile" or "agility" somewhere in the paper, but only [14] presents a characterization to describe it. Herein the definition of agility proposed in [4] is used which measures a legged robot's ability to *change its velocity*. Acceleration is defined as the rate of change of velocity, thus the robot's ability to accelerate itself can be used to measure agility. Agile robots can change their velocity abruptly by generating high levels of acceleration.

However, it is often difficult to intuitively understand agility in terms of acceleration capability. Thus the effort here is to build a bridge between acceleration capability analysis and more common measures of motion ability for mobile systems in order to show the correlation between them. The metric considered is the *time to maximum velocity*. The study is performed using an analysis of acceleration capability, referred to as the dynamic capability equations (DCEs), along with simulation techniques in order to determine values for these metrics. A hexapod and the tripod gait are used to illustrate these ideas. This system is chosen because it is simple to generate joint trajectories resulting in the desired locomotion. However, there are some interesting challenges

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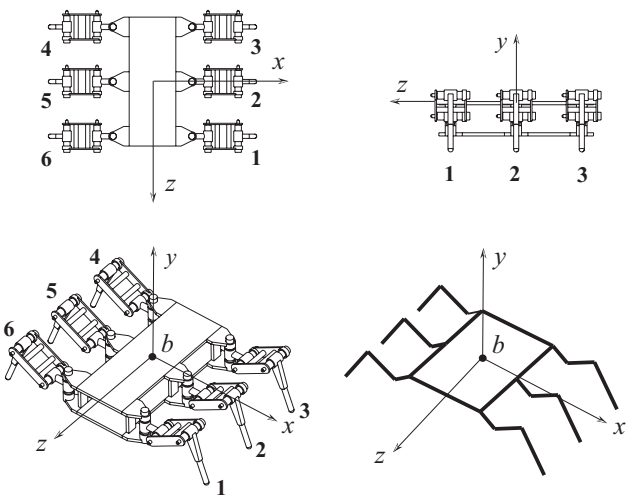


Figure 1: The Hexapod.

involved in actually developing these simulations. These difficulties give one an appreciation for using a performance metric or characterization, such as the DCEs, to aid in the design as opposed to evaluating the metrics using a simulation.

2. THE SCENARIO

The goal of this work is to examine the effects of acceleration capability on locomotion. This effect is investigated in simulation using a model of the hexapod shown in Fig. 1 as a test subject. In these test cases, the hexapod uses the tripod gait to locomote, shown in Fig. 2. This gait is accomplished by placing the odd numbered feet on the ground while lifting the even numbered feet in the air. The odd tripod is then swept forward which moves the body forward. The even tripod is placed on the ground and the odd tripod is lifted off of the ground. The even tripod then sweeps backward and the cycle is repeated. The same cycle can be used for walking straight or turning.

The hexapod's acceleration capability is affected by several different factors including its inertial properties, configuration, ground contact forces, and actuator torque capacity. In order to examine how these factors affect its locomotion, it is necessary to simulate the applied forces. This means that a controller which can generate the tripod gait has to be developed, and the contact forces have to be applied to the dynamic model in a manner that is physically meaningful. These developments created some interesting challenges in assessing how acceleration capability affects the turning radius and achievement of the robot's top speed. The next sections discuss the elements of the simulation in detail.

3. THE SIMULATION

A hybrid dynamic simulation approach was used to deal with the intermittent contact between the hexapod's feet and the ground. The equations of motion for the hexapod can be expressed as

$$A(\mathbf{q})\,\ddot{\mathbf{q}} \;+\; \mathbf{b}(\dot{\mathbf{q}},\mathbf{q}) \;+\; \mathbf{g}(\mathbf{q}) \;=\; \mathbf{\Gamma} \;=\; \mathsf{G}^T\mathbf{\Upsilon} \qquad (1)$$

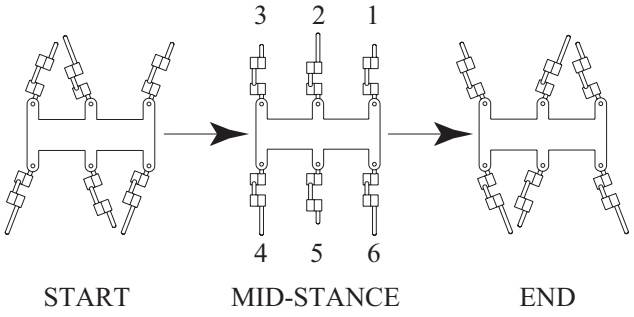


Figure 2: The Tripod Gait.

where $\mathbf{q} \in \mathbb{R}^{24}$ contains the generalized coordinates, and $\dot{\mathbf{q}}$ and $\ddot{\mathbf{q}}$ are its time derivatives of velocity and acceleration. The vectors $\mathbf{b}(\dot{\mathbf{q}},\mathbf{q}) \in \mathbb{R}^{24}$, $\mathbf{g}(\mathbf{q}) \in \mathbb{R}^{24}$, $\mathbf{\Gamma} \in \mathbb{R}^{24}$, $\mathbf{F} \in \mathbb{R}^{c}$, and $\mathbf{\Upsilon} \in \mathbb{R}^{18}$ contain velocity, gravity, actuator torque, external, and actuator forces. The matrices $A(\mathbf{q}) \in \mathbb{R}^{24\times 24}$, $\mathsf{G} \in \mathbb{R}^{24\times 18}$, $J_c \in \mathbb{R}^{24\times c}$ are the mass, actuator transmission, and contact Jacobian matrices; c is the number of contact constraints.

3.1 Contact Forces

When a foot touches the ground, no-slip, no rebound, point contact is assumed. Thus when a foot touches the ground its velocity is set equal to zero. This can be accomplished in several ways, but the details of what was done here are given in [5]. The important thing to note is that (1) does not include contact constraints. Thus they will have to be included in some manner if (1) is used for purposes other than numerical integration.

3.2 Actuator Torques

The computed torque method was used to execute the tripod gait. Since a turning motion is required, the goal is to avoid calculation of the inverse kinematics associated with a turn. Thus an operational space approach was followed. However, the relations in (1) represent an under actuated system. Since the contact constraints are applied during the numerical integration process, a control command which accounts for the effect of the contact constraints is needed.

The constraints are found using a Jacobian of the form

$$\boldsymbol{\vartheta} \;=\; \left[\; \boldsymbol{v}_b^T \;\; \boldsymbol{\omega}_b^T \;\; \boldsymbol{v}_1^T \;\; \cdots \;\; \boldsymbol{v}_6^T \;\right]^T \;=\; J\,\dot{\mathbf{q}} \qquad (2)$$

where $\boldsymbol{v}_b$ and $\boldsymbol{\omega}_b$ are the translational and rotational velocities of the torso and $\boldsymbol{v}_i$ are the translational velocities of the contact point on each foot. Without a loss of generality, assume that the odd tripod is in contact with the ground:

$$\boldsymbol{\vartheta}_f \;=\; \left[\; \boldsymbol{v}_b^T \;\; \boldsymbol{\omega}_b^T \;\; \boldsymbol{v}_2^T \;\; \boldsymbol{v}_4^T \;\; \boldsymbol{v}_6^T \;\right]^T \;=\; J_f\,\dot{\mathbf{q}} \qquad (3)$$

$$\mathbf{0} \;=\; \boldsymbol{\vartheta}_c \;=\; \left[\; \boldsymbol{v}_1^T \;\; \boldsymbol{v}_3^T \;\; \boldsymbol{v}_5^T \;\right]^T = J_c\,\dot{\mathbf{q}} = Q\,R\,E^T\,\dot{\mathbf{q}} \qquad (4)$$

$$= \; Q\,[\;R_c \;\; R_f\;]\,E^T\dot{\mathbf{q}} \;=\; Q\,(R_c\,\dot{\mathbf{q}}_c \;+\; R_f\,\dot{\mathbf{q}}_f) \qquad (5)$$

$$\mathbf{0} \;=\; J_c\,\ddot{\mathbf{q}} \;+\; \dot{J}_c\,\dot{\mathbf{q}} \qquad (6)$$

where $Q \in \mathbb{R}^{24\times 24}$ and $R \in \mathbb{R}^{24\times c}$ are obtained from the QR decomposition of J_c, and $R_c \in \mathbb{R}^{c\times c}$ is upper triangular and full rank. Using (5) yields

$$\dot{\mathbf{q}} \;=\; E\begin{bmatrix} -R_c^{-1}R_f \\ \mathrm{I} \end{bmatrix} \dot{\mathbf{q}}_f \;=\; P\,\dot{\mathbf{q}}_f \tag{7}$$

$$\ddot{\mathbf{q}} \;=\; P\,\ddot{\mathbf{q}}_f + E\begin{bmatrix} -R_c^{-1}Q^T\dot{J}_c\,\dot{\mathbf{q}} \\ 0 \end{bmatrix} = P\ddot{\mathbf{q}}_f + \dot{P}\dot{J}_c\dot{\mathbf{q}} \tag{8}$$

Since the feet stick to the ground and do not rebound we know that $\boldsymbol{\vartheta}_c = \dot{\boldsymbol{\vartheta}}_c = \mathbf{0}$ so $\boldsymbol{\vartheta}_f$ is used to define the operational space:

$$\boldsymbol{\vartheta}_f \;=\; J_f\,\dot{\mathbf{q}} \;=\; J_f P\,\dot{\mathbf{q}}_f \tag{9}$$

$$\dot{\boldsymbol{\vartheta}}_f \;=\; J_f\,\ddot{\mathbf{q}} + \dot{J}_f\dot{\mathbf{q}} \;=\; J_f P\ddot{\mathbf{q}}_f + J_f\dot{P}\dot{J}_c\dot{\mathbf{q}} + \dot{J}_f\dot{\mathbf{q}}\,. \tag{10}$$

All of the velocity relationships above have a dual involving static forces which can be expressed as

$$\left(P^TP\right)^{-1}P\,\dot{\mathbf{q}} = \dot{\mathbf{q}}_f \quad\longrightarrow\quad \boldsymbol{\Gamma} = P^T\left(P^TP\right)^{-T}\boldsymbol{\Gamma}_f \tag{11}$$

and from (1)

$$\left(\mathsf{G}\mathsf{G}^T\right)^{-1}\mathsf{G}\,\boldsymbol{\Gamma} \;=\; \boldsymbol{\Upsilon}\,. \tag{12}$$

The relations in (7)-(12) are used to transform the computed torque method into operational space as

$$\dot{\boldsymbol{\vartheta}}_d + K_v\left(\boldsymbol{\vartheta}_d - \boldsymbol{\vartheta}_f\right) + K_p\left(\mathbf{x}_d - \mathbf{x}\right) = \boldsymbol{\Gamma}_f^* \tag{13}$$

$$P^T\left(AP\left((J_fP)^{-1}\left(\boldsymbol{\Gamma}_f^* - J_f\dot{P}\dot{J}_c\dot{\mathbf{q}} - \dot{J}_f\dot{\mathbf{q}}\right)\right) + \mathbf{b} + \mathbf{g}\right) = \boldsymbol{\Gamma}_f \tag{14}$$

$$\left(\mathsf{G}\mathsf{G}^T\right)^{-1}\mathsf{G}P\left(P^TP\right)^{-T}\boldsymbol{\Gamma}_f \;=\; \boldsymbol{\Upsilon} \tag{15}$$

where $\dot{\boldsymbol{\vartheta}}_d$, $\boldsymbol{\vartheta}_d$, and $\mathbf{x}_d$ represent the desired trajectories for the torso and the feet; see (3). The terms K_p and K_v are diagonal gain matrices. Note that $\mathbf{x}$ contains a set of Euler angles to represent the arbitrary orientation of the torso. It is possible to find other forms of (15), but the one given is less likely to involve singular matrix products.

The different postures in the tripod gait are considered as repeating via points for the desired trajectory [6]. The via points are connected using cubic splines with zero velocity at the start and end of each segment. The turning motion is obtained by specifying the desired orientation of the torso.

3.3 Performance Analysis

The DCEs are used to analyze the robot's performance. A recent paper gave an extensive discussion of the DCEs for legged robots in contact with the environment [4]. The resulting characterization describes how well a legged robot can use ground contact to accelerate itself. The performance analysis is too involved to fully present here so only the highlights will be given.

The goal is to determine to compare the torque required to achieve desired motions, with the amount of torque available. In these simulations the available torque is determined by the motors and has the form of a set of inequalities

$$\boldsymbol{\Upsilon}_{lo} \;\le\; \boldsymbol{\Upsilon} \;\le\; \boldsymbol{\Upsilon}_{hi}\,. \tag{16}$$

In this work, each element in $\boldsymbol{\Upsilon}_{lo}$ and $\boldsymbol{\Upsilon}_{hi}$ is a constant. After the contact constraints are applied to the equations of motion in (1), they are substituted into (16) to obtain the governing equations for the performance analysis. Further information on this analysis is provided in [4].

The analysis is carried out in terms of *balanced quantities* defined as

$$\dot{\boldsymbol{v}}_b^T\dot{\boldsymbol{v}}_b \;=\; |\dot{v}_b|^2 \qquad\qquad \dot{\boldsymbol{\omega}}_b^T\dot{\boldsymbol{\omega}}_b \;=\; |\dot{\omega}_b|^2 \tag{17}$$

where $|\dot{v}_b|$ is the balanced translational acceleration of the torso. The leftmost relation in (17) represents a sphere with a radius of $|\dot{v}_b|$. Note the difference between the radius $|\dot{v}_b|$ and the magnitude of the vector $\dot{\boldsymbol{v}}_b$ which is $\|\dot{\boldsymbol{v}}_b\|$. The idea is to find the maximum values of the balanced quantities subject to the (1) and (16). For a given configuration and contact state, an analytic solution to this problem exists, and it is the result of the DCE analysis referred to as the dynamic capability hypersurface.

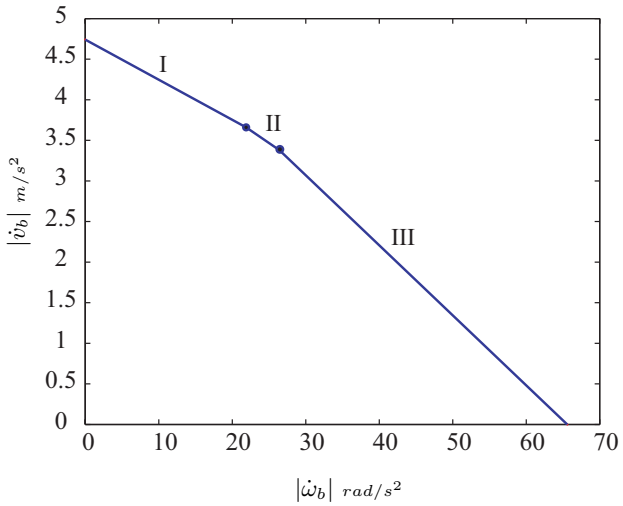


Figure 3: Performance Curve.

A curve from the hypersurface is shown in Fig. 3. It shows the combinations of translational and rotational acceleration of the torso that is guaranteed to be achievable in and about any direction.

A key feature of this analysis is that it can provide the acceleration capability in a particular direction, by defining balanced quantities such as

$$\dot{v}_{b_1}^2 \;=\; |\dot{v}_{b_1}|^2\,. \tag{18}$$

This information is used in the example of Sec. 4.

4. TIME TO VELOCITY

In the automobile industry it is common to quote the time it takes to accelerate from 0mph to 60mph as a measure of performance. This type of measure is also examined for a hexapod with a slightly different interpretation. These test case were run for a set of trajectories which implement the tripod gait. The trajectories for the different phases of the gait were encoded as cubic splines with a start and final time to completion. Therefore the amount of time to complete a cycle of the tripod gait was specified.

A fast and a slow tripod gait were used with prescribed cycle times of 0.39s and 1.2s. There were also two actuator types used, a weak and strong one whose peak torques

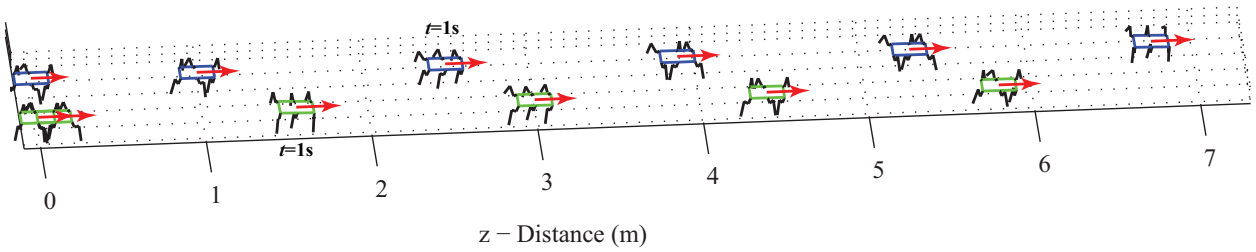


Figure 4: Low Acceleration Capability Hexapods in a 2.5 Second Race. Motion capture is taken at 0.5s intervals. Both hexapods have a maximum torque of 500Nm. The top and bottom hexapods use the fast and slow trajectories. The top hexapod's actuators saturate, limiting its top speed. The time label at $t = 1$s shows the relative position of the hexapods. The length of the arrow extending from each hexapod indicates its acceleration capability in the forward direction; both hexapods have about the same acceleration capability.

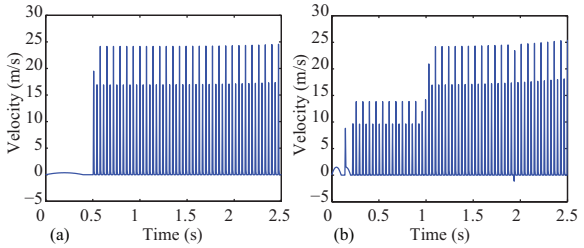


Figure 5: Velocity in the Forward Direction with Peak Torque 500Nm. (a) Hexapod executing the slow trajectories, and (b) the fast trajectories.

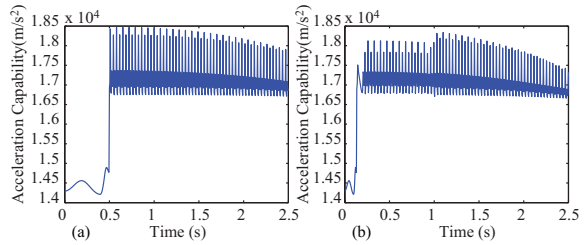


Figure 6: Acceleration Capability in the Forward Direction with Peak Torque 500Nm. (a) Hexapod executing the slow trajectories, and (b) the fast trajectories.

were 500Nm and 1000Nm. If the controller commands more torque than is available, the actuator is clipped to the maximum torque, as would occur in the real world. This clipping prevents the robot from asking for an infinite amount of torque, but also limits its acceleration capability and therefore its speed.

This is illustrated in Fig. 4 where both hexapods have actuators with a peak torque of 500Nm, but the top hexapod is commanded to perform the fast trajectories while the lower one performs the slow ones; the hexapods are identical. The top hexapod moves faster than the lower one but it reaches a maximum speed of about 25m/s as shown in Fig. 5b. The fast trajectories push the hexapod to move faster than this, but its torque limitations prevent it.

Also notice in Fig. 5 that the speed varies greatly during the tripod gait, so even if the hexapod has a large amount of acceleration capability, as shown in Fig. 6, it cannot utilize all of it all of the time due to the trajectories. Both hexapods have roughly the same amount of acceleration capability in the forward direction, as indicated by the arrows in Fig. 4 and in Fig. 6. The top hexapod has a little less because it achieves higher velocities which create velocity forces that require torque to overcome.

The lower hexapod gets of to a slow start, in Fig. 5a, but quickly approaches its maximum velocity. If it had not had this difficulty starting off, it would have kept pace with the top hexapod. However, notice that the top hexapod also had some difficulty getting started in Fig. 5b.

In Fig. 7 the peak torque of the top hexapod is increased to 1000Nm. The bottom hexapod in Figs. 4 and 7 are identical. The increase in peak torque gives the top hexapod more acceleration capability thereby allowing it to achieve the higher speeds required to execute the fast trajectories. Comparing Figs. 4 and 7 notice that the top hexapod moves farther in the 2.5 second run time. Fig. 8b shows that the top hexapod did achieve a higher velocity 35m/s, as compared to the top speed of 25m/s in Fig. 5b. Indeed its acceleration capability in the forward direction was also larger, as shown in Fig. 9b, than in the earlier race, Fig. 6b.

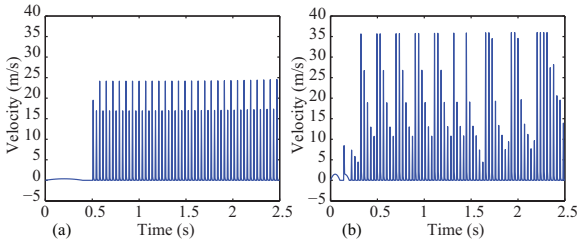


Figure 8: Velocity in the Forward Direction. (a) Hexapod executing the slow trajectories with peak torque of 500Nm, and (b) the fast trajectories with peak torque 1000Nm.

These two cases show how increasing the acceleration ca-

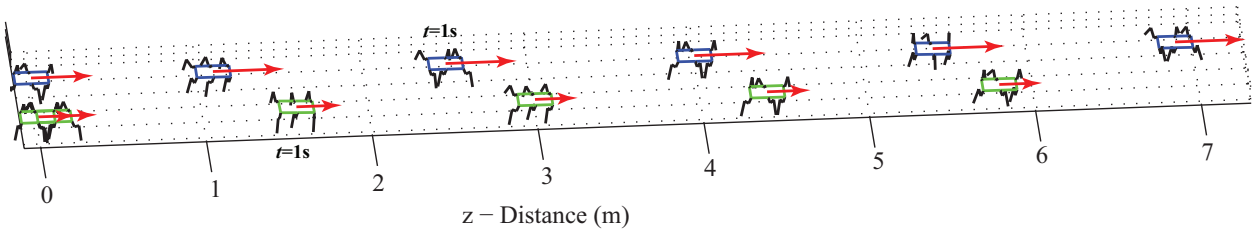


Figure 7: High vs. Low Acceleration Capability Hexapod in a 2.5 Second Race. Motion capture is taken at 0.5s intervals. The top hexapod has maximum torque of 1000Nm while the bottom one has 500Nm. The higher acceleration capability allows the top robot to move farther in 2.5 seconds than in the previous race. The time label at $t = 1$s shows the relative position of the hexapods. The length of the arrow extending from each hexapod indicates its acceleration capability in the forward direction; the longer arrows show that the fast hexapod has more acceleration capability.

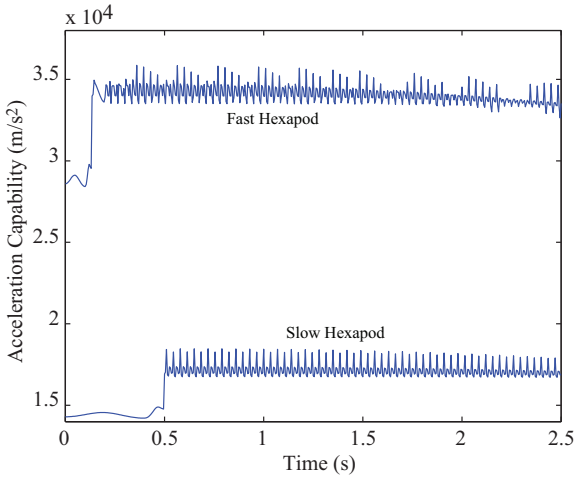


Figure 9: Acceleration Capability in the Forward Direction. (a) Hexapod executing the slow trajectories with peak torque of 500Nm, and (b) the fast trajectories with peak torque 1000Nm.

pability can improve the performance of a hexapod. The original hexapod can accelerate from 0-25m/s in 2.5s, but after an increase in acceleration capability it could accelerate from 0-35 in 2.5s. Here the increase in acceleration capability is due to increased actuator size. It is also possible that the pattern of locomotion could be changed to increase its acceleration capability to achieve the same effect.

What should be noticeable is the number of factors that come into play in the simulation which affect the results. Clearly a change in gait pattern would affect the performance. A performance metric which requires a simulation to evaluate it would have to be run each time any of these factors changes in order to evaluate the performance. This is not a simple task when considering legged robots. It is actually simpler and faster to use the analytic solution encoded in the DCE to design in the desired performance. A simulation will most likely be needed in the end to check the final conceptual design, but it need not slow the design process down when a performance measure is available.

5. CONCLUSIONS

This paper showed that acceleration capability has a significant impact on a commonly used performance metric, the time to a particular speed, which allows it to be used in design in lieu of a simulation. This is advantageous because of the difficulty involved in simulating the locomotion of legged robots. There are several factors which will affect the simulation results, which are more easily addressed using a performance characterization like the DCE. Future work involves examining more agile motions like turning in order to consider rotational performance.

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